Equivalence relations are effective: the quotient identification type is the relation

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'lem:sets_exact' (source lines 339-345, Lemma 10.1.8); the proof environment starts on the next line after the statement, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 10, section 1, theorem offset 7) make the counted lemma print as Lemma 10.1.8, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: equivalence relations are effective (Lemma 10.1.8).

Allowed standard prerequisites (editorial).

- Set-quotient of an equivalence relation. For an equivalence relation (A,R) the quotient A/R is the set-quotient and c_R : A -> A/R is the quotient map.. It is the object on which the identity type is analysed.
- Mere propositions form a set. The universe of mere propositions is a set, by univalence.. It is what allows the double induction defining the extended relation at source lines 349-350.
- Encode-decode method. To identify a family with an identity type one defines a code family, a decoding map, and proves the two composites are identities.. It is the structure of the proof, and the author names it at source line 360.

Lemma 10.1.8. *Suppose (A, R) is an equivalence relation. Then there is an equivalence*

$$(c_R(x) = c_R(y)) \simeq R(x, y)$$

for any $x, y : A$. In other words, equivalence relations are effective.

Proof. We begin by extending R to a relation $\widetilde{R} : A/R \to A/R \to \mathsf{Prop}$, which we will then show is equivalent to the identity type on A/R. We define $\widetilde{R}$ by double induction on A/R (note that Prop is a set by univalence for mere

propositions). We define $\widetilde{R}(c_R(x), c_R(y)) :\equiv R(x, y)$. For $r : R(x, x')$ and $s : R(y, y')$, the transitivity and symmetry of R gives an equivalence from $R(x, y)$ to $R(x', y')$. This completes the definition of $\widetilde{R}$.

It remains to show that $\widetilde{R}(w, w') \simeq (w = w')$ for every $w, w' : A/R$. The direction $(w = w') \to \widetilde{R}(w, w')$ follows by transport once we show that $\widetilde{R}$ is reflexive, which is an easy induction. The other direction $\widetilde{R}(w, w') \to (w = w')$ is a mere proposition, so since $c_R : A \to A/R$ is surjective, it suffices to assume that w and w' are of the form $c_R(x)$ and $c_R(y)$. But in this case, we have the canonical map $\widetilde{R}(c_R(x), c_R(y)) :\equiv R(x, y) \to (c_R(x) = c_R(y))$. (Note again the appearance of the encode-decode method.) □